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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: Understanding Biology

Lesson	Title (MINESEC progression)
1	Epistemology of Biology
2	Presenting biological information and safety rules
3	Scientific thinking and processes
4	Biostatistics and biological data collection
5	Biological data analysis and bio-economics
6	Scientific communication in biology (Bionews)
7	Biology equipment and principles of microscopy
8	The light microscope: structure, use and specimen preparation

Lesson 1: Epistemology of Biology

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define biology and name its main branches
- ✔ list the levels of biological organisation in order
- ✔ distinguish between a hypothesis, a theory and a law

Lesson notes

Biology is the scientific study of living organisms: their structure, functions, growth, origin, evolution and distribution. **Epistemology** asks how we come to know things; in biology, knowledge comes from careful observation, experiment and logical reasoning.

- **Nature of biology:** it is an experimental science; its knowledge is tentative (it can change when new evidence is found); it is based on evidence that others can test.
- **Scope:** biology has many branches, such as botany (plants), zoology (animals), microbiology (microorganisms), anatomy (structure), physiology (functions), genetics (inheritance), ecology (organisms and their environment), biochemistry and biotechnology.
- **Validity:** a biological claim is valid if it is supported by reliable, repeatable evidence obtained by a fair method.
- **Limitations:** biology cannot answer moral or religious questions; living systems are very variable, so results always have some uncertainty; some experiments are limited by ethics (e.g. on humans).
- **Applications:** medicine and health, agriculture and food production, conservation, forensic science, industry (brewing, drugs, biofuels).

KEY POINTS

Hierarchy of biological organisation: atom → molecule → organelle → cell → tissue → organ → organ system → organism → population → community → ecosystem → biosphere

A **hypothesis** is a testable explanation. A **theory** is a well-tested explanation supported by much evidence (e.g. cell theory, theory of evolution). A **law** describes a pattern that always holds under stated conditions (e.g. Mendel's laws of inheritance).

- **Biology and other sciences:** biology uses chemistry (biomolecules, reactions), physics (light in photosynthesis, nerve impulses, microscopy), mathematics (statistics, growth) and geography (distribution of organisms, climate).

Worked examples

Example: Distinguish between a hypothesis and a theory, giving one example of a biological theory.

A hypothesis is a proposed, testable explanation for an observation, which may be supported or rejected by experiment.

A theory is a broad explanation that has been tested many times and is supported by a large body of evidence.

Example: the cell theory (all living things are made of cells).

Practice Questions: Lesson 1

1. Arrange in order of increasing complexity: organ, cell, ecosystem, tissue, population, organism.
2. Name the branch of biology that studies (a) tissues (b) heredity (c) the interaction of organisms with their environment.
3. Distinguish between a hypothesis and a theory.

Answers: *1. cell, tissue, organ, organism, population, ecosystem 2. (a) histology (b) genetics (c) ecology 3. a hypothesis is a testable explanation; a theory is an explanation supported by much tested evidence*

Lesson 2: Presenting biological information and safety rules

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ make clear, labelled biological drawings and calculate their magnification
- ✔ present data in tables and graphs
- ✔ state the main laboratory safety rules

Lesson notes

Presenting information: biological data and observations are presented as tables, graphs, drawings, flow charts and written reports.

- **Biological drawings:** use a sharp pencil; continuous clean lines (no shading or sketching); draw large (at least half a page); label lines ruled, not crossing, ending on the structure; give a title and the magnification (or scale).

KEY POINTS

Magnification = size of drawing (image) ÷ actual size of the specimen

Safety in the laboratory:

- Wear a laboratory coat and safety goggles; tie back long hair; never eat or drink in the laboratory.
- Know the safety icons: toxic (skull and crossbones), corrosive (liquid eating a hand), flammable (flame), harmful/irritant (exclamation mark), biohazard (three linked crescents), oxidising (flame over a circle).
- Report all accidents and spills to the teacher; wash hands after practical work.

Field study safety: work in groups, tell someone where you are going, wear suitable shoes and clothes, avoid dangerous animals and plants, carry a first-aid kit.

Working with organisms: wash hands after handling soil, plants or animals; treat animals humanely and return them to their habitat; never culture microorganisms at 37 °C (human body temperature) in school, because pathogens may grow.

Worked examples

Example: A student's drawing of a leaf is 12 cm long. The real leaf is 4 cm long. Calculate the magnification.

Magnification = 12/4 = × 3

Practice Questions: Lesson 2

1. A drawing of an insect is 9 cm long; the insect is 1.5 cm long. Calculate the magnification of the drawing.
2. State four rules for a good biological drawing.
3. State three laboratory safety rules.

Answers: *1. × 6 2. sharp pencil; clean continuous lines, no shading; ruled label lines not crossing; title and magnification 3. e.g. no eating or drinking; wear goggles and a lab coat; report breakages and spills; wash hands after practical work*

Lesson 3: Scientific thinking and processes

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the steps of the scientific method
- ✔ design a simple controlled experiment

Lesson notes

Steps of the scientific approach (scientific method):

- 1. **Observation**: notice something and ask a question.
- 2. **Hypothesis**: suggest a testable explanation.
- 3. **Prediction**: state what should happen if the hypothesis is correct.
- 4. **Experiment**: test the prediction, changing only the independent variable.
- 5. **Results**: record data carefully (tables, graphs).
- 6. **Conclusion**: decide whether the results support the hypothesis.
- 7. **Communication**: share the results so others can repeat the work.

Designing a good research method: include a **control** (for comparison), keep all other variables constant (controlled variables), use a large enough sample, repeat the experiment (reliability) and measure accurately.

Shortcomings of the scientific approach: results can be affected by bias or human error; samples may be too small; some variables cannot be controlled (especially in the field); a hypothesis can be supported but never proved absolutely.

Worked examples

Example: A student wants to find out whether fertiliser increases the growth of maize. Outline a suitable experiment.

Hypothesis: maize plants given fertiliser grow taller than plants without fertiliser.

Use two groups of 20 maize seedlings of the same variety and age, in the same soil, pots, light and watering.

Test group: add a measured amount of fertiliser; control group: no fertiliser.

Measure the height of each plant every week for 6 weeks; calculate the mean height of each group.

Compare the means (and use a statistical test) to draw a conclusion.

Practice Questions: Lesson 3

1. State the purpose of a control in an experiment.
2. Why should an experiment be repeated several times?

Answers: *1. to show that the result is caused by the factor being tested 2. to make the results reliable and to reduce the effect of chance or errors*

Lesson 4: Biostatistics and biological data collection

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ identify independent, dependent and controlled variables
- ✔ describe methods of collecting biological data

Lesson notes

Biostatistics is the use of statistics to collect, analyse and interpret biological data.

- **Independent variable:** the factor the investigator changes (e.g. temperature).
- **Dependent variable:** the factor that is measured (e.g. rate of enzyme reaction).
- **Controlled variables:** factors kept constant (e.g. pH, volume of enzyme).

Types of data:

Type	Description	Example
Qualitative	descriptions, not numbers	colour of flowers
Quantitative discrete	counted whole numbers	number of leaves
Quantitative continuous	measured, any value in a range	height, mass
Primary data	collected by the investigator	measurements in an experiment
Secondary data	collected by others	published records, journals

Tools and methods of collection: rulers, balances, thermometers, measuring cylinders, quadrats, questionnaires and interviews, direct observation, sampling (random, systematic, stratified), photographs and data loggers.

Worked examples

Example: In an investigation of the effect of light intensity on the rate of photosynthesis in pondweed, identify the independent, dependent and two controlled variables.

Independent: light intensity (distance of the lamp).

Dependent: rate of photosynthesis (number of bubbles of oxygen per minute).

Controlled: temperature, carbon dioxide concentration (amount of $NaHCO_3$), same piece of pondweed.

Practice Questions: Lesson 4

1. In an experiment on the effect of temperature on the rate of yeast respiration, name the independent and dependent variables and two controlled variables.
2. Distinguish between qualitative and quantitative data.

Answers: *1. independent: temperature; dependent: rate of CO_2 production; controlled: e.g. mass of yeast, volume and concentration of sugar solution, pH 2. qualitative: descriptive (colour, shape); quantitative: numerical measurements*

Lesson 5: Biological data analysis and bio-economics

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ calculate measures of central tendency and spread
- ✔ explain what a test of significance shows
- ✔ carry out simple bio-economic calculations

Lesson notes

KEY POINTS

Mean = sum of values ÷ number of values; median = middle value when arranged in order; mode = most frequent value

Standard deviation (s) measures how spread out the data are around the mean: $s = \sqrt{[\Sigma(x - \bar{x})^2/(n - 1)]}$

Tests of significance (e.g. t-test, chi-squared test) show whether a difference is likely to be real or due to chance. If $p < 0.05$, the difference is significant.

- **Data representation:** line graphs (continuous data, e.g. growth with time), bar charts (categories), histograms (frequency of continuous data), pie charts (proportions), scatter graphs (relationship between two variables).
- **Correlation:** a relationship between two variables (positive, negative or none). Correlation does not prove causation. **Regression** fits a line of best fit to make predictions.

Worked examples

Example: The lengths (cm) of 5 leaves are 6, 8, 7, 9 and 10. Calculate the mean and standard deviation.

Mean = 40/5 = 8 cm

Deviations: −2, 0, −1, 1, 2; squares: 4, 0, 1, 1, 4; sum = 10

$s = \sqrt{(10/4)} = \sqrt{2.5} = 1.58$ cm

Bio-economics applies economic thinking to biological activities such as farming, fishing and conservation (costs, income, profit).

Example: A farmer spends 150 000 FCFA on seeds, fertiliser and labour for a plot of tomatoes and sells 600 kg at 400 FCFA per kg. Calculate the profit and the percentage profit.

Income = 600 × 400 = 240 000 FCFA

Profit = 240 000 − 150 000 = 90 000 FCFA

Percentage profit = 90 000/150 000 × 100 = 60%

Practice Questions: Lesson 5

1. Find the mean, median and range of: 12, 15, 11, 18, 14.
2. A poultry farmer spends 400 000 FCFA and sells 250 chickens at 2 500 FCFA each. Calculate the profit and percentage profit.

Answers: *1. mean 14; median 14; range 7 2. 225 000 FCFA; 56.25%*

Lesson 6: Scientific communication in biology (Bionews)

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ communicate biological information clearly to different audiences
- ✔ plan a simple health education campaign

Lesson notes

Scientists share their findings so that others can check, repeat and build on them.

- **Journals:** peer-reviewed articles with a title, abstract, introduction, method, results, discussion and references.
- **Posters and charts:** show key findings visually at conferences, in schools and health centres.
- **Talks and presentations:** oral reports, often with slides.
- **Slogans and health education:** short memorable messages (e.g. "Sleep under a treated mosquito net every night").
- **Educational campaigns:** radio, television, social media, community meetings.

EXAM TIP

Good scientific communication is clear, accurate, honest, uses the correct units and gives the source of information.

Worked examples

Example: Design a short slogan and suggest two methods to educate a village community about cholera prevention.

> Slogan: "Boil it, cook it, peel it or leave it!"
>
> Methods: a poster at the market and health centre showing how cholera spreads; a talk at a community meeting with a demonstration of hand washing and water treatment.

Practice Questions: Lesson 6

1. Suggest three ways to inform a rural community about malaria prevention.

Answers: *1. e.g. talks at community meetings or churches, posters in local languages, radio programmes, demonstrations of treated nets*

Lesson 7: Biology equipment and principles of microscopy

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ identify common biology equipment and its uses
- ✔ define magnification and resolution and convert between units
- ✔ compare light and electron microscopes

Lesson notes

Equipment	Use
Hand lens	magnify small specimens (× 5 to × 20)
Light microscope	view cells and tissues (up to about × 1500)
Petri dish	culture microorganisms, hold small specimens
Pooter	collect small insects
Quadrat	sample plants in an area
Pitfall trap	catch crawling animals
Sweep net	catch insects in grass
Potometer	measure rate of water uptake by a shoot
Respirometer	measure rate of respiration

KEY POINTS

Magnification is how many times larger the image is than the object. **Resolution** (resolving power) is the ability to distinguish two points close together as separate.

Actual size = image size ÷ magnification; 1 mm = 1000 µm; 1 µm = 1000 nm

Feature	Light microscope	Electron microscope
Radiation used	light	beam of electrons
Maximum magnification	about × 1500	over × 500 000
Resolution	about 200 nm	about 0.2 nm
Specimens	living or dead	dead only (in a vacuum)
Image	coloured	black and white (false colour may be added)
Cost and size	cheap, portable	expensive, large

- **Transmission electron microscope (TEM):** electrons pass through a thin section; shows internal structures.
- **Scanning electron microscope (SEM):** electrons bounce off the surface; gives 3-D images of surfaces.

Worked examples

Example: A mitochondrion is 5 mm long in an electron micrograph with a magnification of × 2500. Calculate its actual length in µm.

Actual length = 5 mm ÷ 2500 = 0.002 mm = 2 µm

Practice Questions: Lesson 7

1. A cell is 40 mm wide in a photograph magnified × 800. Find its actual width in µm.
2. Give two differences between a light microscope and an electron microscope.

Answers: *1. 50 µm 2. e.g. the electron microscope has a much higher resolution (about 0.5 nm vs 200 nm) and magnification; it uses electron beams in a vacuum, so only dead specimens can be viewed*

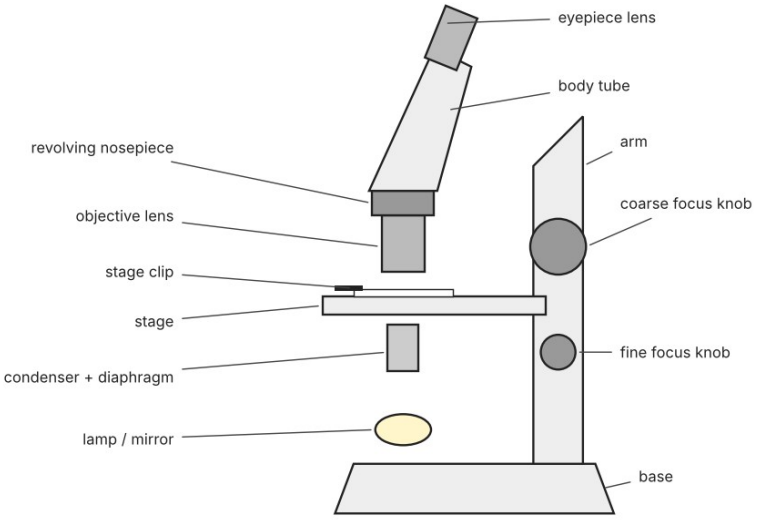
Lesson 8: The light microscope: structure, use and specimen preparation

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ name the parts of the light microscope and their functions
- ✔ prepare and stain a temporary slide

Lesson notes



Part	Function
Eyepiece lens	magnifies the image (e.g. × 10)
Objective lenses	magnify the specimen (× 4, × 10, × 40)
Stage and clips	support and hold the slide
Condenser and diaphragm	focus and control the amount of light
Coarse focus knob	large movements to bring the specimen into focus
Fine focus knob	sharp focusing at high power
Mirror / lamp	provides light

KEY POINTS

Total magnification = eyepiece magnification × objective magnification (e.g. 10 × 40 = × 400)

Using the microscope:

- 1. Carry it with both hands; place it on a firm bench.
- 2. Start with the lowest power objective; adjust the light.
- 3. Place the slide on the stage, with the specimen over the hole.
- 4. Looking from the side, lower the objective close to the slide with the coarse knob.
- 5. Looking through the eyepiece, focus by moving the objective UP (away from the slide).
- 6. Use the fine knob for sharp focus; change to higher power and refocus with the fine knob only.

Preparing a temporary slide (onion epidermis): peel a thin layer of epidermis; place it flat in a drop of water or iodine solution on a clean slide; lower a cover slip at an angle with a mounted needle to avoid air bubbles; remove excess liquid with filter paper.

Stains: iodine (stains starch and nuclei of plant cells), methylene blue (animal cells such as cheek cells), eosin, safranin.

Worked examples

Example: Why should a cover slip be lowered at an angle?

To push out air and avoid trapping air bubbles, which would hide parts of the specimen.

Practice Questions: Lesson 8

1. A microscope has a × 10 eyepiece and a × 40 objective. What is the total magnification?
2. Name a stain used for (a) onion epidermis (b) starch.

Answers: *1. × 400 2. (a) iodine solution or methylene blue (b) iodine solution*

Exam-Style Questions: Understanding Biology

Example: Describe the steps of the scientific method, using an example.

1. Observation: plants near a window grow towards it.
2. Hypothesis: shoots grow towards light.
3. Experiment: grow seedlings with light from one side (test) and all round (control); keep other variables the same.
4. Results and conclusion: only the test seedlings bend, so the hypothesis is supported.
5. Repeat to check the results are reliable, then report them.

Example: An image of a cell is 50 mm long. The magnification is × 2000. Find the actual length of the cell in µm.

Actual size = image size/magnification = 50 mm/2000 = 0.025 mm = 25 µm

Review Exercise: Understanding Biology

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. List the levels of biological organisation from cell to biosphere.
2. Give four differences between the light microscope and the electron microscope.
3. A cell measures 40 mm in a photograph magnified × 2000. Calculate its actual size in µm.
4. The heights (cm) of 6 seedlings are 4, 6, 5, 7, 3 and 5. Calculate the mean, median and mode.
5. What is the purpose of a control in an experiment?
6. State three safety rules for field work.

Answers: *1. cell → tissue → organ → organ system → organism → population → community → ecosystem → biosphere 2. see the table in Lesson 7 3. 20 µm 4. mean 5 cm, median 5 cm, mode 5 cm 5. to provide a standard for comparison, so that any change can be shown to be caused by the independent variable 6. e.g. work in groups, wear suitable boots and clothing, carry a first-aid kit, tell an adult where you are going*

END OF FREE SAMPLE

Get the full notes

You have just read the opening of **Lower Sixth Biology**. The complete book has every lesson of the MINESEC progression sheet, in order:

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